

PALLE MOHAN KRISHNA

COMPUTER SCIENCE ENGINEER | AI & ML

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Professional Summary

Computer Science graduate specializing in Artificial Intelligence and Machine Learning, with hands-on experience in Python, Machine Learning, NLP, Deep Learning, Generative AI, and Retrieval-Augmented Generation (RAG). Developed AI/ML applications using LangChain, FAISS, Hugging Face Transformers, and Streamlit, along with embedded and IoT systems using ESP32, Arduino, and Embedded C. Strong foundation in Data Structures and Algorithms, SQL, Computer Vision, and data processing, with experience building practical projects focused on intelligent automation and real-world problem solving.

Education

B. Tech in Computer Science and Engineering (AI & ML)

Nov 2022 - May 2026

Rajeev Gandhi Memorial College of Engineering and Technology

Nandyal, AP

- CGPA: 8.0/10.0

Technical Skills

Programming Languages	Python, C, Java, SQL, Embedded C
AI/ML & GenAI	Machine Learning, NLP, Deep Learning, Computer Vision, LLMs, RAG, LangChain, Hugging Face
Frameworks & Libraries	Streamlit, Sentence Transformers, NumPy, Pandas, Scikit-learn
Tools & Platforms	Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Arduino IDE
Embedded Systems	Arduino, ESP32, Microcontroller Programming
Core Concepts	Data Structures & Algorithms, OOP, Data Processing, Document Chunking, Text Embeddings

Projects

Smart Campus Assistant using Multi-Source Retrieval-Augmented Generation (RAG)

Technologies: Python | LangChain | FAISS | ChromaDB | Hugging Face Transformers | Streamlit

- Developed a RAG-based Smart Campus Assistant using Python, LangChain, FAISS, and Hugging Face to answer student queries from multiple documents.
- Implemented semantic and hybrid search, document chunking, embeddings, conversational memory, and a Streamlit interface for context-aware responses.

Smart Walking Stick using ESP32 and Ultrasonic Sensors

Technologies: ESP32 | Embedded C | GPS | Bluetooth | Ultrasonic Sensors | Soil Moisture Sensor

- Developed an ESP32-based smart walking stick using ultrasonic and soil moisture sensors to detect obstacles, water surfaces, and manholes.
- Integrated Bluetooth voice feedback, GPS emergency location sharing, and real-time sensor processing for safer navigation.

Flight Passenger Satisfaction Prediction

Technologies: Python | Scikit-learn | Pandas | NumPy | Streamlit

- Developed a Random Forest model to predict airline passenger satisfaction using data preprocessing, feature engineering, ETL, and statistical modeling.
- Built an interactive Streamlit application for real-time prediction and evaluated model performance using classification metrics.

Autonomous Obstacle Avoidance Robot

Technologies: Arduino Uno | Embedded C | Ultrasonic Sensor | Servo Motor | L298N

- Designed an autonomous obstacle avoidance robot using Arduino Uno, ultrasonic sensors, and servo-controlled scanning for real-time detection.
- Integrated the L298N motor driver, distance measurement, and embedded control logic for smooth, collision-free navigation.

Fastest Line Following Robot

Technologies: Arduino Uno | Embedded C | IR Sensors | PID Controller

- Developed a high-speed line-following robot using Arduino Uno, IR sensors, and PID control for accurate real-time path tracking.
- Optimized PID parameters and motor speed control to reduce oscillations and improve stability through sharp curves.

CERTIFICATIONS

- AWS Certified Cloud Practitioner – Amazon Web Services (AWS)

ACHIEVEMENTS & ACTIVITIES

- 1st Prize – Codethon, Rajeev Gandhi Memorial College of Engineering and Technology Nandyal.
- 1st Prize – Ardubotics Hackathon, ECE Department, RGM CET, Nandyal.
- Participant – Technoxian World Robotics Championship 9.0, robotics design, programming, and automation.